Indian Statistical Institute
M. Tech. (CS) I

Machine Learning 1

End Term, Date: March 04, 2024

Timing: 2 hrs. 30 mins | Maximum marks: 60

Attempt ALL questions. The maximum marks you can obtain is 60. Use of a calculator is permitted.

1. Consider finding a weight vector $w \in \mathbb{R}$ that minimizes the cost function $J(w) = w^T Aw + \overset{T}{b}w + \alpha$, where $A \in \mathbb{R}^{n\times n}$ is a symmetric matrix, $b \in \mathbb{R}^n$ is a vector, and α is a scalar.
   (a) Derive an expression for $\nabla J(w)$ and a closed-form expression for a critical point w^* of $J(w)$. For full points, your expression for w^* should be a critical point even when A is singular, assuming that J has a critical point.
   (b) Write the gradient descent update rule for this optimization problem, with step size ϵ.
   (c) We define the "error" in an iterate $w^{(k)}$ to be $e^{(k)} = w^{(k)} - w^*$. Rewrite your update rule in the form "$e^{(k+1)} \leftarrow$ some function of $e^{(k)}$", so that $e^{(k)}$ appears only once and the letters w and b do not appear at all.
   (d) Show that if every eigenvalue λ_i of A satisfies $0 < \lambda_i < 1$, then each successive error $e^{(k+1)}$ is shorter than the previous error $e^{(k)}$, so the iterations cause the error to converge to zero.

   (3+2+4+5 = 14)

2. (a) Suppose we are reliability testing n units taken randomly from a population of identical appliances. We want to estimate the mean failure time of the population. We assume the failure times come from an exponential distribution with parameter $\lambda > 0$, whose probability density function is $f(x) = \lambda e^{-\lambda x}$(on the domain $x \geq 0$).
   (i) In an ideal (but impractical) scenario, we run the units until they all fail. The failure times are $t_1, t_2, ..., t_n$. Formulate the likelihood function $L(\lambda; t_1, ..., t_n)$ for our data. Then find the maximum likelihood estimate $\hat{\lambda}$ for the distribution's parameter.
   (ii) In a more realistic scenario, we run the units for a fixed time T. We observe r unit failures, where $0 \leq r \leq n$, and there are $n - r$ units that survive the entire time T without failing. The failure times are $t_1, t_2, ..., t_r$. Formulate the likelihood function $\mathcal{L}(\lambda; n, r, t_1, ..., t_r)$ for our data. Then find the maximum likelihood estimate $\hat{\lambda}$ for the distribution's parameter.

   (b) Consider the following matrix:

$$X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & -1 \end{bmatrix}$$

$Y = X\theta.$
$\theta = (X^TX)^{-1}X^TY$

   (i) One way to compute the SVD by hand is to exploit the relationship between the singular values of X and the eigenvalues of a related symmetric matrix. What symmetric matrix has eigenvalues related to the singular values of X? Compute that matrix, write down its characteristic polynomial and simplify it. Then determine its eigenvalues. Show your work!
   (ii) Explicitly write down enough eigenvectors of your symmetric matrix to form a complete basis for the space. (Hint: the symmetry of your matrix should make it easy to guess eigenvectors; then, you can confirm that they are eigenvectors.) Based on these, write down the value of U or V (whichever is appropriate).

   (3+5+3+4 = 15)

3. Consider this sample of six points $X_i \in \mathbb{R}^2$:

$$\begin{pmatrix}0\\0\end{pmatrix}, \begin{pmatrix}0\\1\end{pmatrix}, \begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}1\\2\end{pmatrix}, \begin{pmatrix}2\\1\end{pmatrix}; \begin{pmatrix}2\\2\end{pmatrix}.$$

(a) Compute the mean of the sample points and write the centered design matrix $\hat{X}$.

(b) Find all the principal components of this sample. Write them as unit vectors.

(c)
- Which of those two principal components would be preferred if you use only one?
- What information does the PCA algorithm use to decide that one principal components is better than another?
- From an optimization point of view, why do we prefer that one?

(d) Compute the vector projection of each of the original sample points (not the centered sample points) onto your preferred principal component. By "vector projection" we mean that the projected points are still in $\mathbb{R}^2$. (Don't just give us the principal coordinate; give us the projected point.)

(2+3+3+4 = 12)

4. (a) You are given a design matrix $X \in \mathcal{R}^{n\times d}$, storing n training points with d features, and a vector $y \in \mathcal{R}^n$ of continuously-valued labels. As usual, let X_i be training point i, expressed as a column vector such that row i of X is X_i^T. We want to perform least-squares regression with the quadratic regression function $h(z) = z^T W z$ (no linear nor constant terms), where $W \in \mathcal{R}^{d\times d}$ is symmetric. The objective is to find the symmetric matrix W that minimizes the least-squares cost function.

- Derive an expression for $\nabla_w J(w)$.
- Suppose that X has rank n (that is, the training points are linearly independent and $n \le d$). Let I_i be the vector that is row i of the $n \times n$ identity matrix; hence, its entries are zero except a 1 at position i. Observe that we can write training point i as the row vector $X_i^T = I_i X$. Let X^+ be the Moore–Penrose pseudoinverse of X. Write the labels as a diagonal $n \times n$ matrix $Y = diag(y_i)$. Show that the cost function $J(W)$ is minimized by $W^* = X^+ Y (X^+)^T$.

4 wX - y R

(b) Recall that the objective function for L2-regularized linear regression is $J(w) = ||Xw - y||^2 + \lambda||w||^2$. Consider running Newton's method to minimize J. Let w_0 be an arbitrary initial guess for Newton's method. Show that w_1, the value of the weights after one Newton step, is equal to w^*, the global minimizer of J.

(2+4+6 = 12)

5. Recall the loss function for k-means clustering with k clusters, sample points $x_1, ..., x_n$, and centers $\mu_1, ..., \mu_k$:

$$J = \sum_{j=1}^{k} \sum_{x_i \in S_j} ||x_i - \mu_j||^2,$$

where S_j refers to the set of data points that are closer to μ_j than to any other cluster mean.

(a) Instead of updating μ_j by computing the mean, let's minimize L with batch gradient descent while holding the sets S_j fixed. Derive the update formula for the first cluster center μ_1 with learning rate (step size) ϵ.

(b) Recall that in the update step of the standard algorithm, we assign each cluster center to be the mean (centroid) of the data points closest to that center. It turns out that a

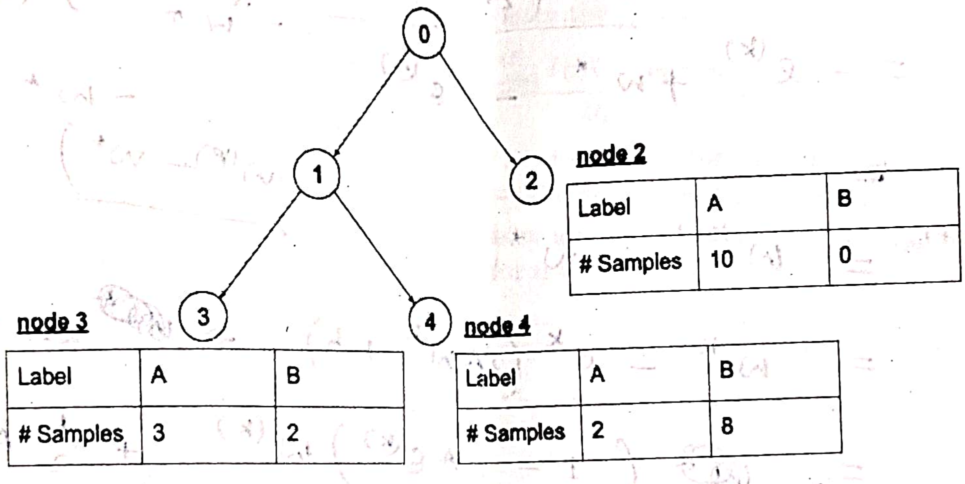
$-y^T X w$

$X^T y$

particular choice of the learning rate ϵ (which may be different for each cluster) makes the two algorithms (batch gradient descent and the standard k-means algorithm) have identical update steps. Let's focus on the update for the first cluster, with center μ_1. Calculate the value of ϵ so that both algorithms perform the same update for μ_1.

(3+5 = 8)

6. The decision tree drawn below stores sample points from two classes, A and B. The tables indicate the number of sample points of each class stored in each leaf node.



node 2

Label	A	B
# Samples	10	0

node 3

Label	A	B
# Samples	3	2

node 4

Label	A	B
# Samples	2	8

Figure 1: A decision tree

(a) What is entropy at the root (Treenode 0)?

(b) What is the information gain of the split at the root node (Treenode 0)? Show your work so we understand how you got that answer.

(3+3 = 6)

7. Given the 2-d data for two classes:

$$\omega_1 = [\begin{pmatrix}1\\1\end{pmatrix}, \begin{pmatrix}1\\2\end{pmatrix}, \begin{pmatrix}1\\4\end{pmatrix}, \begin{pmatrix}2\\1\end{pmatrix}, \begin{pmatrix}3\\1\end{pmatrix}, \begin{pmatrix}3\\3\end{pmatrix}],$$

$$\omega_2 = [\begin{pmatrix}2\\2\end{pmatrix}, \begin{pmatrix}3\\2\end{pmatrix}, \begin{pmatrix}3\\4\end{pmatrix}, \begin{pmatrix}5\\1\end{pmatrix}, \begin{pmatrix}5\\4\end{pmatrix}, \begin{pmatrix}5\\5\end{pmatrix}].$$

Determine the optimal projection line in a single dimension following the Fisher's linear discriminant method. Show all your steps. (8)

$(M_1 - M_2)$

$u^T A u$

Best of luck!

No of A = 15

$En_{root}(D) = \frac{10}{25}$ En Node2 +

$D = En = -\frac{15}{25}\log\frac{15}{25} - \frac{10}{25}\log\frac{10}{25}$

0	
Node2	A A A A
Node1	A B } 5A, A B } 10B

$$w^{(k+1)} = w^{(k)} - \varepsilon^{(k)} \nabla_w J(w^{(k)})$$
$$= w^{(k)} - \varepsilon^{(k)} (2A w^{(k)} + b)$$

∇^2 $2A$

$w^{(k)} - w^*$

$w^{(k+1)} - w^{(k)} =$ $\quad 0 < \lambda_i < 1/2$ $\quad w^{(k)} = w^* + e^{(k)}$

$=$

$$e^{k+1} = w^{(k+1)} - w^* \quad e^{(k+1)}$$

$e^{(k)}, A,$

$$= \cancel{w^k - \varepsilon^{(k)}(2Aw^{(k)} + b)}$$

$e^{(k+1)} - w^* = f(e^{(k)}, A, \varepsilon^{(k)})$

$$= \cancel{e^{(k)}} + w^* - \varepsilon^{(k)}$$

$-w^*$

$(w^{(k)} - w^*)$ $\quad w^*(\quad)$

$$e^{k+1} = w^{k+1} - w^*$$
$$= w^k - \varepsilon^k (2Aw^k + b) \quad 2A + \frac{1}{2}A^{-1}b.$$
$$= (I - 2A\varepsilon^{(k)})\, w^{(k)} + \frac{1}{2}A^{-1}b - 2\varepsilon^k b$$

$$x^{(k+1)} = (I - \eta A)\, x^{(k)}$$

$w^* = \frac{1}{2}A^{-1}b$

$w^{(k+1)} =$

$(\frac{1}{2}A^{-1} + \varepsilon^{(k)})\, b$

$$= (I - \varepsilon^{(k)} 2A)\, w^{(k)} - \varepsilon^{(k)} b.$$

$$(I - 2\varepsilon^{(k)} A)\, w^*$$
$$= w^* - 2\varepsilon^{(k)} A \cdot \frac{1}{2}A^{-1}b$$

Indian Statistical Institute
M. Tech. (CS) I
Machine Learning 1
End Term, Date: May 07, 2024

Timing: 3 hrs. 15 mins | Maximum marks: 100

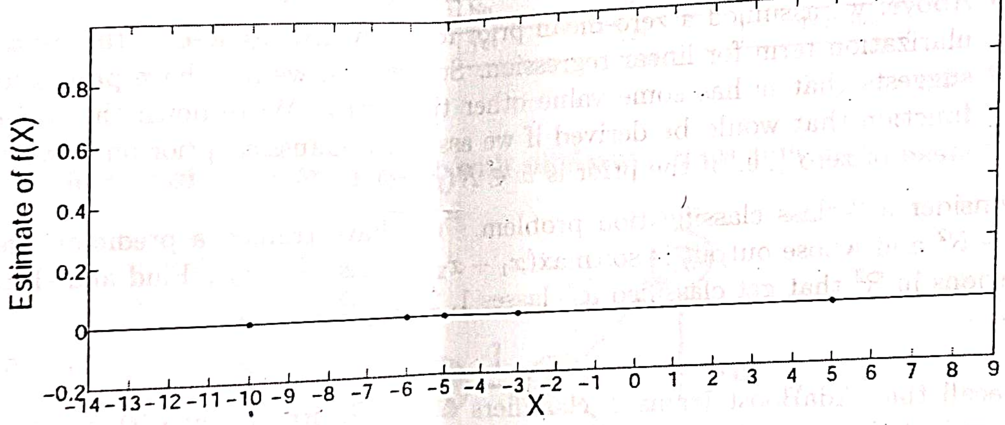
Attempt ALL questions. The maximum marks you can obtain is 100. Use of a calculator is permitted.

1. (a) Consider kernel density estimation using a boxcar kernel. In other words, given a sample of observed data $x_1, ..., x_N$, we will estimate the probability density function as:

$$\hat{f}(X) = \frac{\sum_{j=1}^{N} I(|X - x_j| \leq h)}{2hN},$$

where $|.|$ indicates absolute value and $I(.)$ is an indicator function that takes on value 1 when the logical statement inside the parenthesis is true and 0 otherwise.

- In the figure below, each dot represents a sampled data point. In your answer script, draw a similar figure showing $\hat{f}(x)$ for $h = 3$.



0.6.9 / 1000
0.0069

- Let $F(X)$ represent the true cumulative distribution function for a random variable X and let $f(X)$ represent the probability density function for this same variable. Show that:

$$\mathbb{E}(\hat{f}(X)) = \frac{F(x+h) - F(x-h)}{2h}.$$

(b) Derive the 2×2 real symmetric matrix whose eigenvalues are 5 and 2, such that $(2, -1)^T$ is an eigenvector with eigenvalue 5.

(c) Consider the two-dimensional bivariate normal distribution $N(0, \Sigma)$ where the covariance matrix Σ is the matrix you derived in part (b) and the mean is $\mu = 0$. Let $f(x)$ be the PDF of that normal distribution, where $x \in \mathbb{R}^2$. What are the lengths of the major and minor axes of the ellipse?

$$f(x) = \frac{1}{4\pi\sqrt{10}}.$$

$4a - 2b = 20$
$a + 2b = 1$

(3+6+4+5 = 18)

2. (a) Consider a simple linear regression model in which y is the sum of a deterministic linear function of x and random noise ϵ, as follows:

$$y = wx + \epsilon,$$

$4a - 2b = -20$
$a - 2b = 6$

$2a + b = 10$
$2a - 4b = 12$
$-5b = -2$
$b = -4/5$

$\frac{1}{1}\ [^{1}\ a$

$4a + 2b = 20$
$a - 2b = 6$
$5a = 26$

$3.5\ \frac{26}{9}$

where x is the real-valued input, y is the real-valued output, and w is a single real-valued parameter to be learned. Here ϵ is a real-valued random variable that represents noise, and that follows a Gaussian distribution with mean 0 and standard deviation σ; that is, $\epsilon \sim N(0, \sigma)$.

- Note that y is a random variable because it is the sum of a deterministic function of x, plus the random variable ϵ. Write down an expression for the probability distribution governing y, in terms of $N()$, ϵ, w, and x.
- You are given n i.i.d. training examples $(x_1, y_1), (x_2, y_2), ..., (x_n, y_n)$ to train this model. Let $\mathcal{Y} = (y_1, ..., y_n)$ and $\mathcal{X} = (x_1, ..., x_n)$, write an expression for the conditional data likelihood: $p(\mathcal{Y}|\mathcal{X}, w)$.
- Here you will derive the expression for obtaining a MAP estimate of w from the training data. Assume a Gaussian prior over w with mean 0 and standard deviation τ (i.e., $w \sim N(0, \tau)$). Show that finding the MAP estimate w^* is equivalent to solving the following optimization problem:

$$w^* = \text{argmin}_w \frac{1}{2}\sum_{i=1}^{n}(y_i - wx_i)^2 + \frac{\lambda}{2}||w||^2.$$

  Also, express the regularization parameter λ in terms of σ and τ.
- Above, we assumed a zero-mean prior for w, which resulted in the usual $\frac{\lambda}{2}||w||^2$ regularization term for linear regression. Sometimes, we may have prior knowledge that suggests that w has some value other than zero. Write down the revised objective function that would be derived if we assume a Gaussian prior on w with mean μ instead of zero (i.e., if the prior is $w \in N(\mu, \tau)$).

(b) Consider a 3-class classification problem. You have trained a predictor whose input is $x \in \mathbb{R}^2$ and whose output is softmax$(x_1 + x_2 - 1, 2x_1 + 3, x_2)$. Find and sketch the three regions in $\mathbb{R}^2$ that get classified as classes 1, 2, and 3.

(2+3+5+4+4 = 18)

3. (a) Recall that AdaBoost trains T classifiers $G_1, ..., G_T$ iteratively with the training points re-weighted in each iteration. Given a test point z, the final metalearner's output is a linear combination of these learners,

$$M_T(z) = \sum_{t=1}^{T} \beta_t \hat{G}_t(z).$$

AdaBoost trains the metalearner with an exponential loss function. In this problem, we explore another common boosting algorithm known as Gradient Boosting. Let $\rho = M_T(z)$ be a prediction made by the metalearner, and let $l \in \{-1, +1\}$ be the real-world label for the same point z. Gradient Boosting replaces the exponential loss function with the squared error loss:

$$L(\rho, l) = (\rho - l)^2.$$

We will classify into two classes. As usual, assume we are given n training points $X_1, X_2, ..., X_d \in \mathbb{R}^d$ and a vector $y \in \mathbb{R}^n$ of labels with $y_i \in \{-1, +1\}$.

- Give an expression for the total empirical risk of the Gradient Boosting metalearner with a squared error loss function. Express your answer in terms of the β_t's, G_t's, and y_i's.

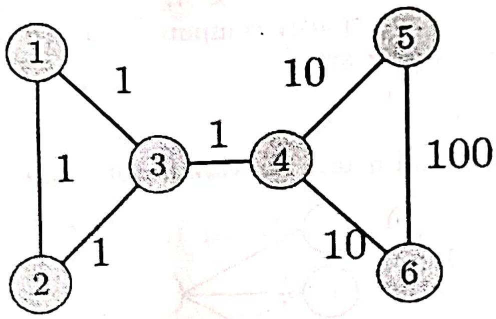
- Suppose Gradient Boosting is in iteration T, and it has just trained a weak learner G_T but it has not yet determined the optimal coefficient β_T. We assume that $\beta_1, ..., \beta_{T-1}$ are already fixed constants that can't be changed during iteration T. Derive an expression for β_T that minimizes the metalearner's empirical risk. Show your work and simplify your answer as much as possible.

(b) Consider a naive Bayes classifier trained on the dataset given below. A new patient comes with x=[**High Fever and Body Ache, but NO runny nose, and NO throat pain**]. Calculate only the posterior probability of having Flu given these symptoms without considering the denominator (P(symptoms)).

	Fever	Body Ache	Runny Nose	Throat Pain	Disease
1	High	Yes	No	Yes	Flu
2	High	Yes	No	No	Flu
3	High	No	Yes	No	Flu
4	Medium	Yes	No	No	Flu
5	Medium	No	No	No	Flu
6	High	Yes	No	Yes	Common cold
7	Low	No	Yes	Yes	Common cold
8	Low	No	Yes	Yes	Common cold
9	Low	Yes	No	No	Common cold
10	Medium	No	Yes	Yes	Common cold

(3+5+7 = 15)

4. (a) Consider the following similarity graph with similarity values annotated on edges.



- Consider the following cuts into two groups S_i and T_i:
  (1) **cut 1:** $S_1 = \{1,2,3\}$, $T_1 = \{4,5,6\}$
  (2) **cut 2:** $S_2 = \{1\}$, $T_2 = \{2,3,4,5,6\}$
  (3) **cut 3:** $S_3 = \{1,2,3,4\}$, $T_3 = \{5,6\}$
  Now compute:
  – the cut weights: $W(S_i, T_i)$.
  – the ratio cut: $\frac{W(S_i,T_i)}{|S_i|} + \frac{W(S_i,T_i)}{|T_i|}$, and
  – the normalized cut: $\frac{W(S_i,T_i)}{vol(S_i)} + \frac{W(S_i,T_i)}{vol(T_i)}$,
  for each of the cuts ($i = 1, 2, 3$). How do the cuts rank from smallest to largest according to each of the cut criterion?

(b) Consider the normalized graph Laplacian $\mathcal{L}$ associated to weight matrix $W \in \mathbb{R}^{d \times d}$ and the degree matrix of the graph being D. Prove that $\mathcal{L}$ is always positive semidefinite. What is the null space of $\mathcal{L}$?

(c) Given that (x_1, x_2) are jointly normally distributed with $\mu = [\mu_1, \mu_2]^T$ and $\Sigma = \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{21} \end{pmatrix}$ with $\sigma_{12} = \sigma_{21}$, derive an expression for the mean of the conditional distribution $p(x_1|x_2 = a)$.

(6+(5+3)+6 = 20)

5. (a) Let $f : C \to \mathbb{R}$ be a differentiable convex function on the convex set C with an L-Lipschitz continuous first-order derivative. Prove that:

$$\frac{1}{2L}\|\nabla f(y) - \nabla f(x)\| \leq f(y) - f(x) - \langle \nabla f(x), y - x \rangle \leq \frac{L}{2}\|y - x\|^2.$$

(b) Suppose your job is to design a multilayer perceptron that receives three binary-valued (i.e., 0 or 1) inputs x_1, x_2, x_3, and outputs 1 if precisely two of the inputs are 1, and outputs 0 otherwise. All of the units use a hard threshold activation function: $\phi(z) = \mathbb{1}(z \geq 0)$. Specify weights and biases of a 3-input 1-output neural network that has one hidden layer with two nodes. Justify your answer.

(c) Suppose you want to train the following model using gradient descent. Here, the input x and target t are both scalar-valued.

$w_{11} +$ $\quad W_{11} + W_{21} \geq 0$

$$z = w_0 + w_1 x + w_2 x^2,$$
$$y = 1 + e^z,$$
$$\mathcal{L} = \frac{1}{2}(\log y - \log t)^2.$$

$f_X(x) = \int_{-\infty}^{x} f(x,y)\, dy$

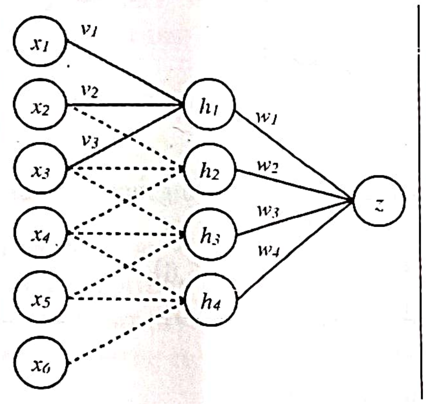
$P(X \leq x)$

Determine the backpropagation rules, which will let you compute the loss derivative $\frac{\partial \mathcal{L}}{\partial w_2}$. Let us denote the partial derivative of $\mathcal{L}$ w.r.t. any variable like y as $\overline{y}$, i.e., $\overline{\mathcal{L}} = 1$. Your equations should refer to previously computed values (e.g., your formula for $\overline{z}$ should be a function of $\overline{y}$). Show all your steps.

(7+7+6 = 20)

6. (a) Consider the convolutional neural network architecture shown in the following figure:

$p(y=1|x) = \frac{p(x|y=1)\,p(y=1)}{p(x)}$



$\begin{bmatrix} 1 & 1 & -2 \\ -1 & 1 & 1 \\ 1 & -1 & 1 \end{bmatrix}$ $\quad -2$

$\int_a^b g'(x)\,dx = [g(x)]_a^b$

In the first layer, we have a one-dimensional convolution with a single filter of size 3 such that $h_i = s(\sum_{j=1}^{3} v_j x_{i+j-1})$. The second layer is fully connected, such that $z = \sum_{i=1}^{4} w_i h_i$. The hidden units' activation function $s(x)$ is the logistic (sigmoid) function. The output unit is linear (no activation function). We perform gradient descent on the loss function $R = (y - z)^2$, where y is the training label for x.

(i) What is this neural network's total number of parameters? Recall that convolutional layers share weights. There are no bias terms.

$y^{(i)}(w^T x^{(i)} + b)$

(ii) Derive an expression for $\frac{\partial R}{\partial w_i}$. Vectorize the previous expression—that is, give an expression for $\frac{\partial R}{\partial w}$.

(iii) Derive an expression for $\frac{\partial R}{\partial v_j}$.

(b) Let $\{(\mathbf{x}_i, y_i)\}_{i=1}^{l}$ be a set of l training pairs of feature vectors and labels. We consider binary classification and assume $y_i \in \{-1, +1\} \forall i$. The following is the primal formulation of L2 SVM, a variant of the standard SVM obtained by squaring the hinge loss:

$$\min_{w, b, \zeta} \ \frac{1}{2}\mathbf{w}^T\mathbf{w} + \frac{C}{2}\sum_{i=1}^{l} \zeta_i^2,$$

$$\text{s.t. } y_1(\mathbf{w}^T\mathbf{x}_i + b) \geq 1 - \zeta_i, \ \ \zeta_i \geq 0, \ \forall i \in \{1, 2, ..., l\}.$$

(i) Show that removing the last set of constraints $\zeta_i \geq 0$, $\forall i$ does not change the optimal solution to the primal problem.

(ii) After removing the last set of constraints, we get a simpler problem:

$$\min_{w, b, \zeta} \ \frac{1}{2}\mathbf{w}^T\mathbf{w} + \frac{C}{2}\sum_{i=1}^{l} \zeta_i^2 \tag{1}$$

$$\text{s.t. } y_1(\mathbf{w}^T\mathbf{x}_i + b) \geq 1 - \zeta_i \ \forall i \in \{1, 2, ..., l\}.$$

Derive the Lagrangian for (1).

(iii) Derive the dual of (1). How is it different from the dual of the standard SVM with the hinge loss?

((2+3+3)+(3+3+7+3)) = 24)

Best of luck!